

STANDARD OPERATING PROCEDURE

SOP: Mechanical Seal Replacement for Pump-101

1. Pre-Requisites and PPE

Prior to arriving at the equipment site, ensure that you have the appropriate tools and replacement OEM-spec single cartridge mechanical seal. The following Personal Protective Equipment (PPE) is strictly required for this task: safety glasses, gloves, and hearing protection.

2. Equipment Isolation

Properly isolate the pump prior to commencing disassembly. Follow the isolation procedure: close the main upstream inlet valve and the downstream outlet valve. Carefully open the designated casing drain and bleed valves to ensure the pump housing is entirely depressurized and drained of all process fluid.

3. Step-by-Step Replacement Instructions

1. Remove the coupling guard and disconnect the coupling between the motor and the pump shaft.
2. Unbolt and carefully remove the bearing housing and impeller assembly from the main casing.
3. Extract the old mechanical seal assembly. Inspect the shaft sleeve for any scoring or damage.
4. Clean the seal chamber thoroughly using approved industrial solvents.
5. Install the new OEM mechanical seal cartridge, ensuring proper alignment with the shaft sleeve.
6. Reassemble the impeller and bearing housing. Tighten all bolts to the manufacturer's specified torque rating.
7. Reconnect the motor coupling and reinstall the coupling guard securely.
8. Close all drain and bleed valves. Slowly open the upstream inlet valve to prime the pump and check for initial static leaks.
9. Once primed, the pump is ready for a test run under operational conditions.